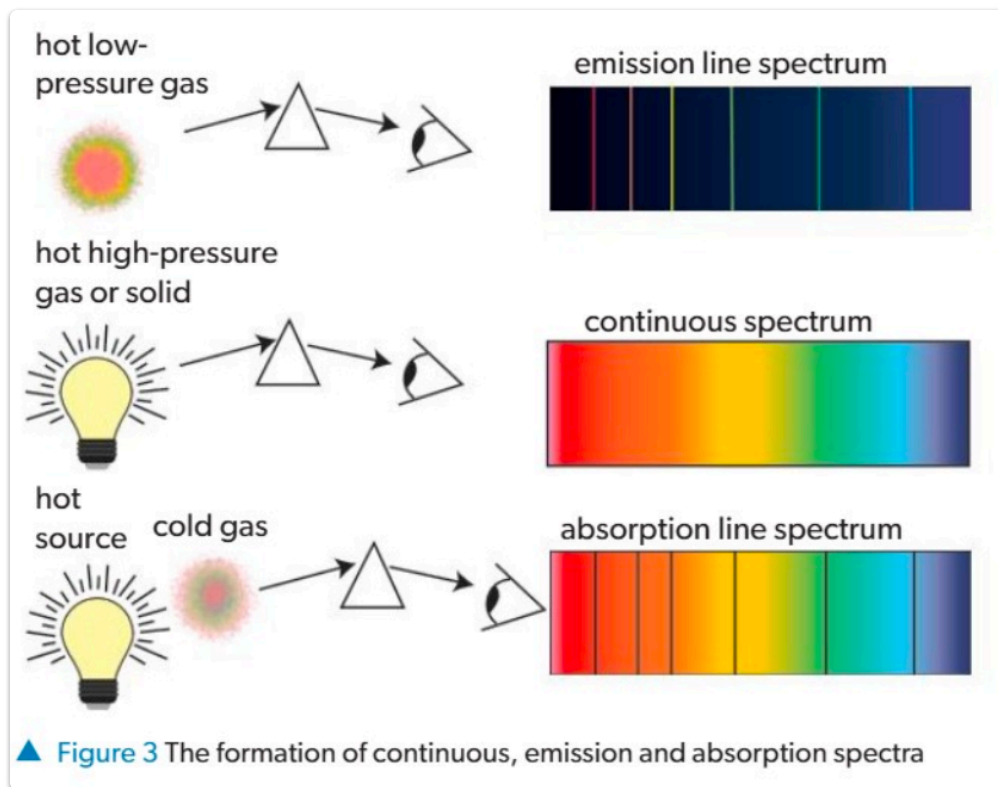


emission spectrum



An **emission spectrum** consists of a series of discrete lines are **precise frequencies** that are produced when **hot, low-pressure** gas atoms **de-excite** from an excited **energy state**.

Individual atoms initially **absorb energy** to go to a **higher energy state**. However, the state is unstable and the atom will drop back to a **lower-energy state**. As it does so, it **transfers the energy** from itself as a quantum of energy.